

Aryan Jeena

US Citizen | aryanj@sas.upenn.edu | (908)-636-5723 | LinkedIn: <https://www.linkedin.com/aryan-jeena/> | GitHub: <https://aryan-jeena.github.io/>

EDUCATION

University of Pennsylvania

Philadelphia, PA

B.S.E in Computer Science and Mathematics, Minor in Statistics, Concentration in Artificial Intelligence

Expected Graduation, May 2028

- o **GPA:** 3.7
- o **Related Coursework:** Programming Languages and Techniques I & II, Mathematical Foundations of Computer Science, Honors Multivariable Calculus III (Parts I & II), Probability, Real Analysis, Discrete Mathematics II, Numerical Optimization for Data Science & Machine Learning, Big Data Analytics, AI Lab: Data, Systems, and Decisions, Intro to Python for Data Science, Statistical Computing with R

EXPERIENCE

Amazon

Redmond, Washington

Software Development Engineer Intern, Project Kuiper

Sept 2026 – Dec 2026

BlackRock

New York, New York

Software Engineering Analyst Intern, Aladdin Engineering Group

Jun 2026 – Aug 2026

Timing

New York, New York

Software Development Intern

May 2025 – Aug 2025

- Designed and implemented smart contact creation modal with autofill from LinkedIn/Google APIs, improving speed-to-input by ~40%.
- Contributed to frontend and backend specs for AI-powered personal CRM, including calendar syncing, reminders, and follow-up classification.
- Created adaptive onboarding logic in TypeScript, using conditional survey inputs to personalize user flows and improve activation by ~25%.
- Developed chatbot functionality using OpenAI Assistants API with real-time message handling, enabling automated networking queries, follow-up suggestions, and calendar-based reminders.

AI in Business Club (AIBC)

Philadelphia, Pennsylvania

Consulting & Development Committee – Proscia Project

Aug 2024 - May 2025

- Collaborated with Proscia, an AI-powered digital pathology company, to develop data-driven solutions for enhancing cancer diagnostics and pathology workflows.
- Conducted machine learning model evaluation for Proscia's Concentriq® platform, analyzing 100,000+ pathology images to improve AI-driven diagnostic accuracy.
- Worked with a team of 5 consultants, leveraging tools such as Python, TensorFlow, and SQL to optimize Proscia's AI models, reducing error rates by ~10–15% in histopathological analysis through hyperparameter tuning and pipeline optimizations.
- Provided strategic recommendations on AI integration and business development, streamlining client adoption processes for leading healthcare institutions and research labs.

Abacus Aviation and Management Services Inc.

Toronto, Canada

Software Engineering Intern

April 2024 - Aug 2024

- Engineered a full-stack inventory management system to digitize parts tracking and maintenance workflows for FAA/EASA-certified aircraft (B747/B787), improving compliance visibility and reducing paperwork by ~50%.
- Designed relational database schema in PostgreSQL to log parts usage, maintenance intervals, and compliance checks; reduced lookup time by 50–60%.
- Built user-facing dashboard in React with real-time status updates, search/filter features, and alert system for expiring parts and service due dates.

PROJECTS

FitMax

Sept 2025 – Oct 2025

- Developed full-stack AI web application integrating FastAPI microservices, Next.js/TypeScript frontend, and PostgreSQL (pgvector) for personalized, data-driven health recommendations via RESTful API architecture.
- Implemented Transformer-based NLP engine using PyTorch and Hugging Face to perform semantic search, vector embeddings, and evidence scoring over PubMed data, powering ML-driven personalization and reducing user query time by ~30%.
- Engineered cloud-ready backend with Docker Compose, Redis caching, and CI/CD automation (GitHub Actions), optimized API latency, and secure JWT authentication.
- Built a responsive analytics dashboard with OpenAPI-generated clients, TailwindCSS, and interactive data visualizations, demonstrating enterprise-grade system design and AI integration.

Soccer Injury Betting

July 2025 – Aug 2025

- Designed and implemented a scalable ML pipeline with modular Python architecture (ingest.py, wrangle.py, model.py), automating ETL workflows via API integration (API-Football), error handling, and validation across 9,000+ records and 1,100+ matches.
- Built predictive regression models using CatBoost and Scikit-learn, applying feature engineering (ELO, rolling averages, differential metrics), hyperparameter tuning, and cross-validation, achieving 18-20% ROI.
- Developed interactive dashboards with Streamlit and Pandas, improving data processing efficiency by ~35% through NumPy vectorization and asynchronous scraping.

Real-Time Portfolio & Risk Management Platform

Aug 2025 – Sept 2025

- Engineered a distributed, event-driven system using FastAPI microservices, Kafka message streaming, and asynchronous I/O to simulate real-time market data ingestion, portfolio tracking, and risk analytics computation with sub-second latency and scalable queue-based architecture.
- Developed fault-tolerant backend services integrating Redis caching, typed SQLAlchemy ORM, and asynchronous database pipelines to compute portfolio KPIs (P/L, volatility, Sharpe ratio, drawdown) and broadcast live updates via WebSockets to a React dashboard.
- Implemented production-grade DevOps and observability stack with Docker Compose, Prometheus-Grafana metrics, structured logging, and CI/CD automation for linting, type checking, and pytest coverage, ensuring reliability, scalability, and maintainability across all services.

SKILLS

Programming: Python, Java, OCaml, TypeScript, JavaScript, SQL, HTML, CSS, FastAPI, React, Next.js, Streamlit, PyTorch, Hugging Face, CatBoost, Scikit-learn

Tools, Frameworks & Analytics: Docker, Redis, Kafka, PostgreSQL, SQLAlchemy, Git, GitHub Actions, Prometheus, Grafana, JWT, WebSockets, Power BI, OpenAPI, CI/CD, Jupyter Notebooks, Eclipse, IntelliJ, VS Code Pandas, NumPy, Tableau, Streamlit, Google Analytics, SEMrush, Microsoft Office Suite